

# CHALMERS



PREVIEW

## Innovation and Intellectual Property

Strategic IP Management and  
Economics of Technology

MARCUS HOLGERSSON

*Department of Technology Management and Economics*  
CHALMERS UNIVERSITY OF TECHNOLOGY  
Gothenburg, Sweden 2012

ProQuest Number:27757836

All rights reserved

INFORMATION TO ALL USERS

The quality of this reproduction is dependent on the quality of the copy submitted.

In the unlikely event that the author did not send a complete manuscript and there are missing pages, these will be noted. Also, if material had to be removed, a note will indicate the deletion.



ProQuest 27757836

Published by ProQuest LLC (2020). Copyright of the Dissertation is held by the Author.

All Rights Reserved.

This work is protected against unauthorized copying under Title 17, United States Code  
Microform Edition © ProQuest LLC.

ProQuest LLC  
789 East Eisenhower Parkway  
P.O. Box 1346  
Ann Arbor, MI 48106 - 1346

PREVIEW

THESIS FOR THE DEGREE OF DOCTOR OF PHILOSOPHY

# Innovation and Intellectual Property

Strategic IP Management and Economics of Technology

MARCUS HOLGERSSON

PREVIEW

Department of Technology Management and Economics

CHALMERS UNIVERSITY OF TECHNOLOGY

Gothenburg, Sweden 2012

Innovation and Intellectual Property  
Strategic IP Management and Economics of Technology  
MARCUS HOLGERSSON

© MARCUS HOLGERSSON, 2012

ISBN 978-91-7385-761-1  
Doktorsavhandlingar vid Chalmers tekniska högskola, Ny serie nr 3442  
ISSN 0346-718X

Industrial Management and Economics  
Department of Technology Management and Economics  
Chalmers University of Technology  
SE-412 96 Gothenburg  
Sweden  
Telephone + 46 (0)31-772 1000

Printed by Chalmers Reproservice  
Gothenburg, Sweden 2012

To Erika and Rune

PREVIEW

PREVIEW

## Abstract

Innovations and technological developments have been recognized for their central importance for economic success and growth at least since the 1930s. Intellectual property (IP) and intellectual property rights (IPRs), such as patents, trade secret rights, and copyrights, have during more recent decades caught increasing attention, and, mainly due to various developments at macro level, IP has become an important source of competitive advantage at micro level in many industries. This has led to an increased importance of strategic IP management, and the related research field has been growing since the late 1990s. This thesis aims to contribute to this growing field, and the first purpose of this thesis is *to explore and explain strategic and innovation related IP management practices, and the managerial and economic consequences of such practices*. Apart from the growing importance of IP management in general, an increased focus on open and collaborative approaches for creating innovations has led to a need for new and adapted IP management skills. The second purpose of this thesis is therefore *to develop managerial and economic frameworks, models, and tools to be used in the intersection between IP management and open innovation practices*. These purposes are addressed in a cover paper and six appended research papers of theoretical/conceptual as well as of empirical nature, being based on interviews, questionnaires, patent statistics, and document studies.

In connection to the first purpose the results show that, while many small firms have problems with properly benefitting from the patent system, large firms have increasingly developed their IP strategies, especially their patent strategies. The purpose is then not only to appropriate monopolistic returns from innovations but also to govern various forms of open innovation. Large firms were found to in a first step increase their patenting (in terms of quantity), and in a second step focus more on selective, quality-oriented, and internationalized patenting. Additional results show that the internationalization leads to a convergence in managerial choices of output markets for patenting worldwide, in parallel with market and technology diversification. Further, a case from mobile telecommunications illustrates the role of IP management in the governance of open innovation systems. Finally, two cases from the automotive industry illustrate the IP-related problems that arise in connection to divestments and other types of disintegrations ('IP disassembly problems'), and how IP management can mitigate them.

This leads to the second purpose, related to the development of models, tools, and frameworks for IP management in relation to open innovation. First, the thesis provides a conceptual framework of innovation openness, especially pinpointing the role of IPRs. This framework emphasizes three key dimensions of innovation openness: resource distribution, technology governance, and technology accessibility. Second, a framework for managing the IP disassembly problem is presented, enabling increased exit opportunities and decreased transaction costs. Third, a method for determining fair, reasonable, and non-discriminatory royalties in licensing collaborations is developed, applicable to multilateral licensing deals.

It stands clear that contemporary IP management is not (and has never been) only about maximizing excludability. Strategic IP management must therefore be developed and integrated with technology and corporate management in order to foster success at the micro level of firms, and thereby also at macro level. Developments in IP management skills (e.g., sourcing, control, commercialization, licensing, valuation, pricing) and IP contracts will then most likely lead to increased efficiency of interorganizational technological relationships and quasi-integrated organizational forms, and thereby also to increased innovativeness and economic development.

**Keywords:** *Intellectual property right; open innovation; research and development; innovation economics; technology management; strategy; value appropriation; licensing; governance; theory of the firm*

PREVIEW

## Acknowledgements

The pursuit of a PhD is a large and difficult task. However, when pursuing it with the support of supervisors, colleagues, friends, and family it is also an enjoyable and highly rewarding undertaking. I have been fortunate to have such support, and I am deeply thankful for that.

Ove Granstrand, who talked me into pursuing a PhD and continued to supervise me along the road, has been of major importance for this thesis. I will always remain grateful for Ove's generosity in terms of ideas, knowledge, personal network, and friendship. Clas Wahlbin, who tragically passed away a little more than a year before the finish of this thesis, made a lasting impact not only on the content of this thesis, but also on me as a person. Clas was genuinely helpful, generous, and kind, with solid methodological insights and experience. Mats Magnusson provided the so necessary push towards finishing the thesis, and contributed both as supervisor and as director of Institute for Management of Innovation and Technology (IMIT).

Feedback has been given by numerous people, and for that I am indebted not only to my supervisors, but also to other colleagues, journal reviewers, and conference discussants. I am especially thankful to Fredrik Tell who gave constructive feedback on a draft of this thesis at the final seminar, leading to a substantial improvement before the final submission. Similarly, the feedback given by Björn Remneland-Wikhamn and Frank Tietze at my licentiate seminar improved an earlier version of this thesis. Thank you!

I am sincerely grateful to the Industrial Management and Economics research group, including MELT project colleagues and external collaborators. This group has provided an inspiring environment with highly knowledgeable scholars. Bengt Domeij and Thomas Ewing have enlightened me about the details and peculiarities of law, and eventually even I came to realize that law is actually quite fun. Multiple discussions with Marcel Bogers and Nicole Ziegler have also been highly rewarding. Similarly, So Young Sohn and her research team at Yonsei University have enabled a stimulating exchange of ideas and contacts.

I also thank all my current and previous colleagues at Chalmers who have supported and inspired me, including everyone in the "innovation corridor". More specifically, I am grateful to all fellow PhD students at Chalmers for exciting discussions about research and other things, and my gratitude goes especially to Anders Billström, Fredrik Wernersson Brodin, Klas Eriksson, and Marcus Linder.

The administrative support at both IMIT and Chalmers has always been careful, efficient, and friendly. I am especially thankful to Birgitta Andersson, Carina Blomkvist, Lucas Hörte, and Bengt Karlsson at IMIT and Eva Burford, Angelica Linnehav, and Yvonne Olausson at Chalmers.

One of the probably most important inputs to the PhD process stems from all the contributing firms, respondents, and interviewees. I am truly grateful for the time you have taken to answer my questions. An equally important input stems from the financial sponsors enabling the research. The financial support from Chalmerska forskningsfonden, E.ON, IMIT, STINT, Vinnova, and Zacco is thus gratefully acknowledged.

Most importantly, I want to express my gratitude to my family. Erika, you are my biggest inspiration. And Rune, who would have thought that a newborn little boy could offer his father so much support? I love you both.

Marcus Holgersson

Gothenburg, October 2012

PREVIEW

## **List of appended papers**

### **Paper I**

The anatomy of rise and fall of patenting and propensity to patent: The case of Sweden. Co-authored with Ove Granstrand. Published in *International Journal of Intellectual Property Management*, 2012, Vol. 5, No. 2, pp. 169-198.

### **Paper II**

Patent management in entrepreneurial SMEs: A literature review and an empirical study of innovation appropriation, patent propensity, and motives. Forthcoming in *R&D Management*.

### **Paper III**

Multinational technology and intellectual property management - Is there global convergence and/or specialization? Co-authored with Ove Granstrand. Forthcoming in *International Journal of Technology Management*.

### **Paper IV**

Conceptualizing innovation openness: A framework and illustrative case. Co-authored with Marcel Bogers and Ove Granstrand. Submitted to an international journal. An earlier version was presented at the R&D Management Conference, 23-25 May, 2012, Grenoble.

### **Paper V**

The 25% rule revisited and a new investment-based method for determining FRAND licensing royalties. Co-authored with Ove Granstrand. Published in *les Nouvelles*, 2012, Vol. 47, No. 3, pp. 188-195.

### **Paper VI**

Managing the intellectual property disassembly problem. Co-authored with Ove Granstrand. Submitted to an international journal. An earlier version was presented at the European Patent Academy workshop at the European Patent Office, 4-5 June, 2012, Munich.

PREVIEW

## List of contents

|          |                                                                                  |           |
|----------|----------------------------------------------------------------------------------|-----------|
| <b>1</b> | <b>INTRODUCTION.....</b>                                                         | <b>1</b>  |
| <b>2</b> | <b>FRAME OF REFERENCE.....</b>                                                   | <b>5</b>  |
| 2.1      | BASIC CONCEPTS.....                                                              | 5         |
| 2.2      | PATENT AND IPR SYSTEMS FOR INCENTIVIZING INNOVATION.....                         | 6         |
| 2.3      | TWO IMPORTANT TRENDS FOR IP MANAGEMENT.....                                      | 9         |
| 2.3.1    | <i>Brief history and the emergence of a pro-patent era.....</i>                  | <i>9</i>  |
| 2.3.2    | <i>Open innovation.....</i>                                                      | <i>15</i> |
| 2.4      | THEORIES OF THE FIRM.....                                                        | 17        |
| 2.4.1    | <i>Transaction cost theory.....</i>                                              | <i>17</i> |
| 2.4.2    | <i>Property rights theory.....</i>                                               | <i>18</i> |
| 2.4.3    | <i>Resource-based theory.....</i>                                                | <i>21</i> |
| 2.4.4    | <i>Firms, hybrids, and markets.....</i>                                          | <i>22</i> |
| 2.5      | INNOVATION APPROPRIATION STRATEGIES.....                                         | 24        |
| 2.6      | STRATEGIC MANAGEMENT OF INTELLECTUAL PROPERTY.....                               | 26        |
| 2.7      | KEY RESEARCH PAPERS WITHIN THE FIELD.....                                        | 30        |
| <b>3</b> | <b>RESEARCH DESIGN AND METHODOLOGY.....</b>                                      | <b>35</b> |
| 3.1      | RESEARCH PROJECTS.....                                                           | 35        |
| 3.2      | RESEARCH PURPOSES AND SUB-STUDIES.....                                           | 35        |
| 3.3      | BASIC ASSUMPTIONS AND RESEARCH STRATEGY.....                                     | 40        |
| 3.4      | RESEARCH METHODS AND DATA SOURCES.....                                           | 41        |
| <b>4</b> | <b>SUMMARIES OF AND CONTRIBUTIONS TO THE APPENDED PAPERS.....</b>                | <b>47</b> |
| 4.1      | PAPER I.....                                                                     | 47        |
| 4.2      | PAPER II.....                                                                    | 47        |
| 4.3      | PAPER III.....                                                                   | 48        |
| 4.4      | PAPER IV.....                                                                    | 49        |
| 4.5      | PAPER V.....                                                                     | 49        |
| 4.6      | PAPER VI.....                                                                    | 50        |
| <b>5</b> | <b>MAIN RESULTS.....</b>                                                         | <b>51</b> |
| 5.1      | DESCRIPTIVE AND EXPLANATORY RESULTS.....                                         | 51        |
| 5.1.1    | <i>Background.....</i>                                                           | <i>51</i> |
| 5.1.2    | <i>Internationalization of Swedish patenting.....</i>                            | <i>52</i> |
| 5.1.3    | <i>Convergence of international patenting.....</i>                               | <i>55</i> |
| 5.1.4    | <i>A change in patenting strategies.....</i>                                     | <i>58</i> |
| 5.1.5    | <i>IP governance and IP assembly – cases from mobile telecommunications.....</i> | <i>60</i> |
| 5.1.6    | <i>The IP disassembly problem – cases from the automotive industry.....</i>      | <i>61</i> |
| 5.1.7    | <i>Patent management in entrepreneurial SMEs.....</i>                            | <i>63</i> |
| 5.2      | PRESCRIPTIVE RESULTS FOR IP MANAGEMENT IN OPEN INNOVATION.....                   | 65        |
| 5.2.1    | <i>Managing innovation openness.....</i>                                         | <i>65</i> |
| 5.2.2    | <i>Managing the IP disassembly problem.....</i>                                  | <i>67</i> |
| 5.2.3    | <i>Proper IP pricing and value sharing.....</i>                                  | <i>68</i> |
| <b>6</b> | <b>DISCUSSION.....</b>                                                           | <b>71</b> |
| 6.1      | INSTITUTIONS, GOVERNANCE, AND LEARNING.....                                      | 74        |
| 6.2      | IP MANAGEMENT AND OPEN INNOVATION.....                                           | 76        |
| 6.3      | IP MANAGEMENT AND ECONOMIC ORGANIZATION.....                                     | 78        |
| 6.4      | IMPLICATIONS FOR MANAGEMENT AND POLICY.....                                      | 80        |
| 6.5      | DIRECTIONS FOR FUTURE RESEARCH.....                                              | 81        |
| <b>7</b> | <b>CONCLUSIONS.....</b>                                                          | <b>85</b> |
|          | <b>LIST OF REFERENCES.....</b>                                                   | <b>87</b> |

PREVIEW

## List of figures

|                                                                                         |    |
|-----------------------------------------------------------------------------------------|----|
| Figure 2.1 Total number of patent applications worldwide per year, 1985-2010 .....      | 11 |
| Figure 2.2 National patent applications in selected countries per year, 1985-2008 ..... | 14 |
| Figure 2.3 A typology of different types of goods .....                                 | 20 |
| Figure 3.1 Relation between thesis purposes and appended papers .....                   | 37 |
| Figure 5.1 Normalized patent application streams .....                                  | 54 |
| Figure 5.2 The innovation process.....                                                  | 65 |
| Figure 5.3 Framework for innovation openness .....                                      | 66 |
| Figure 5.4 Framework for managing the IP disassembly problem.....                       | 68 |
| Figure 6.1 Elements of strategic management of technological IP .....                   | 72 |

## List of tables

|                                                                                                  |    |
|--------------------------------------------------------------------------------------------------|----|
| Table 2.1 Top ten patentees in terms of granted US utility patents in 1987, 2000, and 2011 ..... | 10 |
| Table 2.2 The world's most highly valued trademarks .....                                        | 12 |
| Table 2.3 Examples of factors related to IP management .....                                     | 30 |
| Table 2.4 The most cited papers as identified by the structured literature review .....          | 32 |
| Table 2.5 The most productive scholars and the most common research outlets (journals).....      | 33 |
| Table 2.6 The most frequently cited references by the research field .....                       | 34 |
| Table 3.1 Paper-specific purposes and research questions .....                                   | 39 |
| Table 3.2 Empirical data collection methods in the different papers .....                        | 42 |
| Table 3.3 List of the most important contextualizing interviews .....                            | 43 |
| Table 5.1 Patent application streams related to Sweden.....                                      | 53 |
| Table 5.2 Statistical results on market convergence and concentration .....                      | 57 |
| Table 5.3 Statistical results on technology convergence and concentration .....                  | 58 |
| Table 5.4 Explanatory factors behind a decrease and/or increase of first filings .....           | 59 |
| Table 5.5 Results on patent management in entrepreneurial SMEs .....                             | 64 |

PREVIEW

## List of abbreviations

|       |                                                                       |       |                                                                    |
|-------|-----------------------------------------------------------------------|-------|--------------------------------------------------------------------|
| AIPPI | International Association for the Protection of Intellectual Property | Korea | Republic of Korea ('South Korea')                                  |
| CAFC  | Court of Appeals for Federal Circuit                                  | M&A   | Merger and acquisition                                             |
| CAPM  | Capital asset pricing model                                           | MAD   | Merger, acquisition, and divestment                                |
| CCC   | Change of control clause                                              | MC    | Marginal cost                                                      |
| CEO   | Chief executive officer                                               | MELT  | Management, economics, law, and technology                         |
| China | People's Republic of China                                            | MNC   | Multinational corporation                                          |
| CIPO  | Chief intellectual property officer                                   | NPE   | Non-producing entity                                               |
| CTO   | Chief technology officer                                              | PCT   | Patent Cooperation Treaty                                          |
| EPO   | European Patent Office                                                | PLC   | Product life cycle                                                 |
| ETSI  | European Telecommunications Standards Institute                       | PRT   | Property rights theory                                             |
| FDI   | Foreign direct investment                                             | PTO   | Patent and trademark office                                        |
| FRAND | Fair, reasonable, and non-discriminatory                              | RBT   | Resource-based theory                                              |
| GDP   | Gross domestic product                                                | ROI   | Return on investment                                               |
| GM    | General Motors                                                        | R&D   | Research and development                                           |
| GPL   | General Public License                                                | SME   | Small or medium-sized enterprise                                   |
| IC    | Intellectual capital                                                  | TCT   | Transaction cost theory                                            |
| ICT   | Information and communication technology                              | TRIPS | Agreement on Trade-Related Aspects of Intellectual Property Rights |
| IP    | Intellectual property                                                 | USPTO | United States Patent and Trademark Office                          |
| IPR   | Intellectual property right                                           | VC    | Venture capital                                                    |
| ISA   | International Searching Authority                                     | VCC   | Volvo Car Corporation                                              |
| JV    | Joint venture                                                         | WIPO  | World Intellectual Property Organization                           |
|       |                                                                       | WTO   | World Trade Organization                                           |

PREVIEW

## 1 Introduction

On August 24, 2012, a federal jury in San Jose, California, awarded Apple a one billion US dollar damage from Samsung, its main competitor in the smartphone industry, due to patent infringement. While Apple's market value rose with roughly 15 billion US dollar after the verdict, the Samsung stock price dropped by 7.5%, leading to a decrease in market value of twelve billion US dollar, probably partly due to the risk of an injunction in connection with the final ruling that was yet to come. The stock of another competitor, Nokia, rose by 6% after the decision, probably because Nokia's smartphones used the Microsoft Windows mobile operating system. The Nokia/Windows ecosystem was expected to be less vulnerable to infringement accusations than Google's Android operating system that was used by Samsung and many others. Expectations were therefore that Nokia would be able to catch some of the market shares that would be lost by Samsung in case of an injunction. Expectations were also that the Nokia/Windows ecosystem would grow in popularity among smartphone manufacturers, which in turn would attract application developers and thereby increase the popularity and utility of the ecosystem as a whole.

The case above is only one out of several recent high level court cases illustrating the role of intellectual property rights (IPRs) for firms and their success. These cases have been frequently reported in various news media during the early 2010s, and the importance of IPRs and intellectual property (IP) for technology-based businesses has thus been increasingly highlighted in recent years. However, these issues are not new. In the late 19th century Alexander Graham Bell's patents were central to the success of his business in relation to competitors, although the Swedish telecommunications firm LM Ericsson's initial success was actually enabled much due to the absence of a Swedish patent in Bell's portfolio. In the emerging aircraft industry in the early 20th century the Wright brothers sued a number of competitors for infringing Wright's patents for aircraft control, arguably curbing the US aircraft industry development to the extent that the US government eventually forced the industry to reach cross-licensing agreements in order not to fall too far behind European competitors.<sup>1</sup> In the 1980s "patent wars" were frequently fought, for instance between Japanese and US firms in the electronics industry and between Procter & Gamble and Kimberly Clark in the diaper industry.

Court cases and patent wars like the ones described above are useful examples of the importance of IP, as they provide instances where IP has major implications

---

<sup>1</sup> See the Manufacturer's Aircraft Association.

for the success of firms. Only a small part of all IPRs are ever subject to court cases, however. This thesis studies strategic IP management more generally, with a focus on IP related to technological innovations. Strategic management of technological IP refers to formulating and executing strategies related to technological IP, including issues such as how to acquire and create IP, how to govern IP, and how to exploit and extract value from (commercialize) IP.

A prerequisite for innovation related IP is the development of new innovations. Much research has highlighted the importance of innovations and technological developments for economic growth and welfare (e.g., Baumol, 2002; Rosenberg, 1982; Rosenberg & Birdzell, 1986; Scherer, 1999; Schumpeter, 1934, 1942; Solow, 1956, 1957). Much research has also covered the area of technology and innovation management (e.g., Burns & Stalker, 1961; Chesbrough, 2003; Pavitt, 1990; Teece, 2009; Trott, 2008; Utterback, 1994). The research area of IPRs has traditionally been treated within the disciplines of law and/or economics (e.g., Arrow, 1962; Domeij, 2003; Romer, 2002), while the research on innovation related IP management has traditionally been scarce (Granstrand, 1999; Hanel, 2006). However, IP management research has been growing since the late 1990s (e.g., Granstrand, 1999; Pisano, 2006; Pisano & Teece, 2007; Reitzig, 2004; Somaya, 2012), much due to the macro level policy driven emergence of ‘pro-patent eras’ with increasing patenting around the world (e.g., Granstrand, 1999; Hall, 2005; Hall & Ziedonis, 2001; Hu, 2010; Hu & Jefferson, 2009; Kortum & Lerner, 1998). This thesis contributes to the growing literature on IP management by its first purpose: *to explore and explain strategic and innovation related intellectual property management practices, and the managerial and economic consequences of such practices.*

An important development during the 2000s, with implications for IP management, has been the growing practice and research of open innovation (e.g., Chesbrough, 2003; Dahlander & Gann, 2010; Laursen & Salter, 2006), referring to innovation activities and processes that cross organizational boundaries. Such activities can include technology trade, licensing, collaborative research and development (R&D), crowdsourcing, acquisitions, divestments, etc. The practice of open innovation sets new requirements for strategic IP management. A common traditional assumption has been that IPRs should be used to maximize excludability (protection) of innovations in order to enable high returns from in-house production and commercialization. Although that view is still valid under many conditions, it must be complemented with a more multifaceted portfolio of strategies and strategy combinations in order to align IP management with the general technology and innovation strategy, be it open or closed, in order to foster firm success. This thesis contributes to this area by its second purpose: *to develop managerial and economic frameworks, models, and tools to be used in the*

*intersection between intellectual property management and open innovation practices.*

The thesis focuses on technical inventions and innovation related intellectual property, albeit in a non-exclusive way. This focus puts patents at the core, since patent systems are specifically designed to promote (technological) innovations by giving the owner of a patent the right to exclude others from commercially exploiting the patented invention. Further, the primary focus is on firm-level IP management, rather than national IPR policies and IPR systems. However, interdependencies between micro and macro levels must be taken into account when studying IP management, since national and international institutions (such as patent laws) govern the available set of strategic options for management. Therefore, implications are derived for both management and policy.

This thesis consists of these cover chapters and six appended papers. The cover part is outlined as follows. The introductory chapter is followed by a frame of reference in chapter 2. The methodology and paper-specific purposes and research questions are motivated and described in chapter 3. This is followed by summaries of the appended papers in chapter 4. Some of the main results from the study are described in chapter 5. Chapter 6 contains a discussion of the results, and finally the main conclusions are summarized in chapter 7.

PREVIEW

## 2 Frame of reference

This frame of reference starts with a section in which a number of basic concepts needed throughout the thesis are defined, and continues with a section explaining the rationale of patent and IPR systems. Two important trends underlying this thesis are described in the subsequent section. These trends include the emergence of a pro-patent era, in which the importance of IP for management has grown, and the increased focus on open innovation, setting new requirements on IP management. After the description of these trends, three different theories of the firm are described. A section on appropriation strategies is then followed by a related section on strategic IP management. Finally, key research papers within the field are presented, as identified by a structured literature review.

### 2.1 Basic concepts

Innovations in general, including developments of useful technical knowledge (technology), i.e., technological innovations, are major factors behind economic developments (Baumol, 2002; Rosenberg, 1982; Rosenberg & Birdzell, 1986; Scherer, 1999; Schumpeter, 1934, 1942; Solow, 1956, 1957). An innovation is then commonly defined as something new that has come to some sort of use, a definition that makes a separation between invention and commercialization activities or processes (Freeman, 1982; Garcia & Calantone, 2002; Granstrand, 1999; Layton, 1977; March, 1991; Schumpeter, 1934). Granstrand (1999, p. 58) defines an *innovation* as a “change in ideas, practices or objects involving some degree of (i) novelty or creation based on human ingenuity and (ii) success in application”. An *invention*, in comparison, can be defined as a “first idea, sketch or contrivance of a new-to-the-world product, process or system, which may or may not be patented” (Freeman, 1982, p. 201). An invention is thus turned into an innovation when the invention comes to its first use (it is commercialized), for instance by being sold the first time (in the case of a product invention) or by being usefully applied in production (in the case of a process invention). Inventions and innovations should not be confused with the concept of *discoveries*, i.e., findings of pre-existing features of nature (Granstrand, 1999). This may, for example, be a law of nature. An invention differs from a discovery in that it is invented by man – hence not existent before being invented. An *imitation* is defined as a close reproduction, copy, or duplication of something once perceived as an invention (*ibid.*). Finally, the European Patent Office (EPO) defines (as of 2012) a *patent* as “a legal title granting its holder the right to prevent third parties from commercially exploiting an invention without authorization”.

The main difference between discoveries on one hand and inventions and innovations on the other hand is that the latter two require an active agent that